

OSCE-II

CVS

CHD

1. Acyanotic Congenital Heart Lesions

- Lesions Resulting in Increased **Volume Load**
 - ✓ Eg. **ASD, VSD, PDA, AV-**canal defect (increased pul. vascularity)
- Lesions Resulting in Increased **Pressure Load**
 - ✓ Pulmonary stenosis, aortic stenosis, coarctation of aorta (normal pulmonary vascularity)

1. Cyanotic Congenital Heart Lesions

- Cyanotic Lesions with **Decreased Pulmonary Blood Flow**
 - ✓ e.g. tricuspid atresia, **TOF**, single ventricle with pulmonary stenosis, pulmonary atresia
- Cyanotic Lesions with **Increased Pulmonary Blood Flow**
 - ✓ e.g. **TGA**, TAPVR, truncus arteriosus.

Cont'd

- DX
 1. cyanotic or acyanotic
 2. CXR
 3. ECG/ Echo
- ❖ Hyperoxia test
 - Administration of 100% O₂ improves cyanosis= resp. problem
 - If not cardiac
- Principle of mx
 - Correct the underlying defect
 - Rx of CHF
 - Prevention and treatment of complication

VSD

VSD

- **Indications for surgical closure of a VSD include:**
 - Patients at any age with large defects in whom clinical symptoms and FTT cannot be controlled medically;
 - Infants b/n “6 and 12 mo” of age with large defects associated with pulmonary Htn.
 - Patients >24 month of age with a $Q_p : Q_s$ ratio greater than 2 : 1.
 - Patients with a supracristal VSD of any size.

NB. Severe PVD nonresponsive to pulmonary vasodilators is a contraindication to closure of a VSD.

Complications

- Infective endocarditis
- Recurrent lung infection
- Heart failure
- Pulmonary HTN
- Acquired pulmonary stenosis
- Aortic valve regurgitation

ASD

AVSD

PDA

- May associate with maternal rubella
- c/f
 - bounding peripheral arterial pulses and a wide pulse pressure
 - continuous murmur
 - Heart failure
- PDA is Seen often in Preterm infants and spontaneous closure occurs in most instances
- Spontaneous closure of the ductus after infancy is extremely rare.

Cont'd

- **Complications**

- **Infective
Endocarditis/Endarteriti
s**
- **CHF**
- **Embolization**
- **Pulmonary HTN**

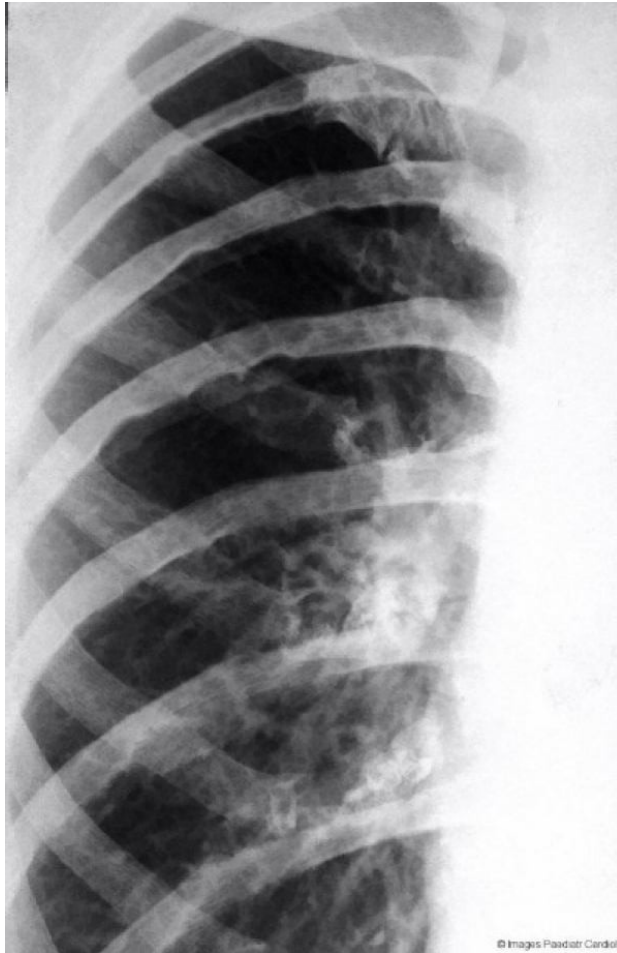
IX

- CXR-normal if PDA is small
- CXR- in large PDA, enlargement of the LA, LV, and PA, and an increase in pulmonary vascular markings .
- ECHO- diagnostic

mx:

- Irrespective of age, patients with PDA require surgical or catheter closure.

COA



- CXR:
 - cardiomegally
 - Pulmonary congestion
 - Notching of the infe. Border of the ribs
- MX
 - prostaglandin E1,
 - surgery

TOF

- CXR
- Boot or wooden shoe shaped heart
- Diminished pulm. BF



Cont'd

- Paroxysmal hypercyanotic attacks (hypoxic, “blue,” or “tet” spells).
- C/f:
 - Increased cyanosis, agitaion, fast and deep breathing, syncope
 - Usually in the morning with initial awakening or after vigorous cry
- Mx:
 1. fluid resuscitation
 2. place the infant in knee chest position
 3. adminster O2
 4. morphine (0.2mg/kg sc)
 5. if acidotic; sodium bicarbonate

Cont'd

- **Complications:**
 - polycythemia
 - Cerebral thromboses
 - Brain abscess
 - Bacterial endocarditis
- **Associated anomalies:**
 - PDA
 - Right aortic arch
 - DiGeorge syndrome
- **Treatment:**
 - Prevent dehydration/polycythemia
 - surgery

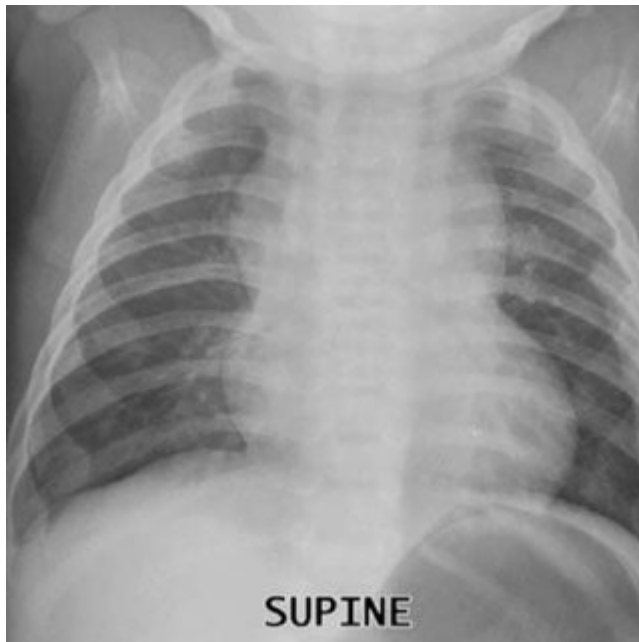
TGA



- CXR- cardiomegally, a narrow mediastinum (“egg-shaped heart”), increased pulmonary blood flow.
- ECHO- is diagnostic
- Treatment:
 - prostaglandin E1
 - Rashkind balloon atrial septostomy
 - Surgery- arterial switch (Jatene) procedure

TAPVR

- CXR: snowman appearance
- Mx:
surgical repair



ARF

Modified Jones Criteria:

- **Major criteria :**

- Carditis
- Arthritis/polyarthralgia
- Sydenham's chorea
- Erythema marginatum
- Subcutaneous nodules

Minor criteria :

- Clinical findings
 - ✓ Arthralgia-
monoarthralgia vs
polyarthralgia
 - ✓ Fever- 38.0C vs 38.50C
- Laboratory findings
 - ✓ Elevated acute phase
reactants (ESR- 30 vs 60,
CRP).
 - ✓ Prolonged PR interval.

Cont'd

- Supporting evidence for antecedent streptococcal pharyngitis:
 - ✓ Positive throat culture or rapid streptococcal antigen test.
 - ✓ Elevated or rising streptococcal antibody titer.
 - antistreptolysin O (80-85%)
 - 95-100% patients have an elevation either of the following -antistreptolysin O, anti-DNase B, antihyaluronidase

- **Diagnosis made with:**

2 major criteria or 1 major and 2 minor criteria

plus

Supporting evidence for antecedent streptococcal pharyngitis (mandatory).

Cont'd

- **Exceptions** (strict adherence to Jones criteria not needed):
 1. Sydenham's Chorea
 2. Indolent Carditis
 3. Rheumatic Fever recurrence

Cont'd

Erythema marginatum



- Evanescent, erythematous, macular nonpruritic rash with pale centers and rounded or serpiginous margins.
- Mostly trunk and proximal extremities.
- accentuated by warming the skin.

Cont'd



Subcutaneous nodules

- Firm, painless, freely movable nodules approximately 1cm size in diameter.
- Most often seen in patients with carditis.
- Usually located over the extensor surfaces of the joints (elbows, knees and wrists), in the occipital portion of the scalp, or over the spinous processes.

Cont'd

- DDX

- Rheumatoid arthritis
- Reactive arthritis
- Malignancy
- SLE
- Viral/bacterial myocarditis/pericarditis

- MX

General

1. Place on bed rest and monitor closely for evidence of carditis.
2. Antibiotic treatment for 10 days with oral penicillin or erythromycin or a single IM dose of Benz. Penicillin.
3. Long-term antibiotic prophylaxis.

Cont'd

- **Anti – inflammatory therapy:**
 - Withheld anti-inflammatory treatment till full blown picture of RF appears.
- **Pain relief** – achieved by acetaminophen.
- **Migratory polyarthritits and carditis with out Cardiomegaly or CHF**
 - → ASA 50-70mg/kg/24hr divided into 4 doses po for 3 – 5 days,
 - then 50mg/kg/24hr for 3weeks and half that dose for 2-4week.
- **Carditis with cardiomegaly or CHF**
 - → Prednisone 2mg/kg/24hr divided into 4 doses po for 2 – 3weeks
 - followed by half the dose for 2-3weeks then taper. While tapering prednisone start ASA 50mg/kg/24hr in 4 divided doses for 6weeks.
- **Supportive treatment**
 - **moderate to severe carditis**
-digoxin, fluid and salt restriction, diuretics, and oxygen.

Cont'd

- Mx: Chorea
 - anti-inflammatory agents are usually not indicated.
 - Sedatives -phenobarbital or haloperidol or chlorpromazine

Cont'd

I. **Primary Prevention** (prompt and proper treatment of GAS pharyngitis) within 9 days of symptom.

a) Benz. Penicillin

wt $\leq$ 27kg $\rightarrow$ 600,000IU IM stat.

wt $>$ 27kg $\rightarrow$ 1,200,000IU IM stat.

b) Penicillin V

Children 250mg po 2-3x/d for 10d.

Adolescents 500mg po 2-3x/d for 10d.

c) Erythromycin (in penicillin allergy)

40mg/kg/24 divided into 2-4 doses po for 10d.

d) Azithromycin (fewer GI side effects)

500mg po on the first day, then 250mg po/d for 4 days.

e) Oral cephalosporins (alternative).

Cont'd

II. Secondary prevention (prevention of recurrence).

- A. Benz. Penicillin 1.2M IU IM every 3 – 4 weeks.
- B. Penicillin V 250mg po bid
- C. Sulfadiazine 500 – 1000mg po/d
- D. Erythromycin 250mg po BID

Table 183-6		Duration of Prophylaxis for People Who Have Had Acute Rheumatic Fever: Recommendations of the American Heart Association	
CATEGORY		DURATION	
Rheumatic fever without carditis		5 yr or until 21 yr of age, whichever is longer	
Rheumatic fever with carditis but without residual heart disease (no valvular disease*)		10 yr or until 21 yr of age, whichever is longer	
Rheumatic fever with carditis and residual heart disease (persistent valvular disease*)		10 yr or until 40 yr of age, whichever is longer; sometimes lifelong prophylaxis	

Infective endocarditis

- **Common:** Native valve or other cardiac lesions
 - Streptococcus Viridans
 - Staphylococcus Aureus
 - Group D Streptococcus
- **Prosthetic valve:**
 - ✓ *Staphylococcus epidermidis*
 - ✓ *Staphylococcus aureus*
 - ✓ Viridans group streptococcus
 - ✓ *Pseudomonas aeruginosa*
- **Highest risk:**
 - prosthetic cardiac valves
 - unrepaired cyanotic congenital heart disease
 - completely repaired defects with **prosthetic material** or device during the 1st 6 mo after repair
 - repaired congenital heart disease with **residual defects** at or adjacent to the site of a prosthetic patch
 - previous infective endocarditis.

Cont'd

- Osler's node
 - Tender pea sized intradermal nodule
 - On the pad of fingers
 - Due to immune depostion



Cont'd



- Janeway lesion
 - Painless small hemorrhagic lesion on palms and soles
 - blanching

Cont'd

- Splinter hemorrhage



Cont'd



- Rose spot
 - White centered retinal hemorrhage

Duke criteria

➤ Major criteria

1. positive blood cultures (2 separate cultures for a usual pathogen, 2 or more for less typical pathogens)
2. evidence of endocarditis on echocardiography
 - Intracardiac mass
 - Regurgitant flow near a prosthesis
 - Abscess
 - Partial dehiscence of prosthetic valves
 - New valve regurgitant flow

➤ Minor criteria

1. predisposing conditions
2. Fever
3. **embolic**-vascular signs
4. **immune complex phenomena** (glomerulonephritis, arthritis, rheumatoid factor, Osler nodes, Roth spots)
5. **Microbiologic**
a single positive blood culture
6. ECHO signs not meeting the major criteria.
6. the presence of newly diagnosed clubbing, splenomegaly, splinter hemorrhages, and petechiae
6. a high ESR and c-reactive protein
6. and the presence of central nonfeeding lines, peripheral lines, and microscopic hematuria

Cont'd

Definite infective endocarditis:

- **A. Pathologic criteria:**
 - 1. Microorganisms demonstrated by culture or histologic examination of a vegetation or
 - 2. Pathologic lesions; vegetation or intracardiac abscess confirmed by histologic examination showing active endocarditis
- **B. Clinical criteria:**
 - 1. Two major criteria or
 - 2. One major criterion and three minor criteria; or
 - 3. Five minor criteria

- **POSSIBLE INFECTIVE ENDOCARDITIS:**

- 1. One major criterion and one minor criterion; or
- 2. Three minor criteria

- **Rejected:**

- 1. Firm alternative diagnosis explaining evidence of IE; or
- 2. Resolution of IE syndrome with antibiotic therapy for <4 days; or
- 3. No pathologic evidence of IE at surgery or autopsy, with antibiotic therapy for <4 days; or
- 4. Does not meet criteria for possible IE as above

Cont'd

- Mx
- Specific :
 - Initial empiric antibiotics to cover Gram positive and negative organisms.
 - Combination of **cloxacillin, ampicillin and gentamycin** for 4-6 weeks (except gentamycin for 2 weeks)
- If there is CHF:
 - diuretics, afterload reducing agents, and in some cases, digitalis.
- Surgery-intractable heart failure, mycotic aneurysm, dehiscence of intracardiac patch

Cont'd

- Complications

- CHF is most common complication
- Systemic/pulmonary emboli
- Mycotic aneurysm
- Heart block

- Prevention

- General dental care and oral hygiene
- Treatment of infection in high risk children
- Antibiotic prophylaxis before
 - surgical procedures of respiratory tract systems

Pericardial disease (acute)

- Infectious
 - Viral : Coxsackie, Echo, EBV, Influenza, HIV
 - Bacterial: TB, staph, hemophilus, pneumococcal, salmonella
 - Fungal/other: histo/ blasto/ coccidio, rickettsia
- Rheumatologic
 - SLE, JRA, Sarcoid,, Dermatomyositis, Ankylosing Spondylitis, Scleroderma, PAN
- Neoplastic
 - Primary: angiosarcoma, mesothelioma
 - Metastatic: breast, lung, lymphoma, melanoma, leukemia
- Immunologic
 - Celiac sprue, Inflammatory Bowel Disease
- Drug
 - Hydralazine, Procainamide
- Other
 - Post Pericardiotomy, Chest Trauma, Aortic dissection, MI, Uremic, Post Radiation
- IDIOPATHIC -86%

types

- Serous pericarditis
- Fibrinous/ serofibrinous
 - Most common type
 - Bread and butter pericarditis
- Purulent/ suppurative
- Hemorrhagic pericarditis
- Caseous pericarditis

Acute pericarditis

c/f

- Often preceding viral illness
1-2wk prior
- Chest Pain
 - Sudden, sharp, pleuritic,
constant, retrosternal
 - worse supine/ L lat decub,
relief in sitting up and **leaning
forward**
 - radiation: back, trapezius
ridge
 - symptoms usually resolve by
2 weeks, ECG abnormalities
may persist for months

- ECG
 - PR depression
 - ST elevation

Cont'd

Management

- NSAIDS/ASA
 - ASA 650 q3-4hr
 - Ibuprofen 300-600 q 6-8 hrs x 1-4days
 - Avoid Indocin, reduces CBF
- Steroids
 - if no response after 48hr NSAID
 - use concurrent NSAID
- Colchicine
 - chronic +/- NSAID
 - useful in recurrent pericarditis
 - symptom free period 3.1 +/- 3mos vs 43 +/- 35mos (p<.00001) in largest multicenter trial to date

- Complications
 - Pericardial Effusion/Tamponade
 - Constrictive Pericarditis
 - can be “transient” – 10% may have transient sx within 1st month, resolves by 3 months
 - Recurrent Pericarditis (20-25%)
 - Rx – NSAIDS/ Colchicine +/- steroids

Constrictive pericarditis

- Cause
 - Cardiac surgery
 - Radiotherapy
 - TB
 - Collagen vascular disease
 - Post viral pericarditis
 - Infiltration by neoplasms
- MX
 - The only effective treatment for chronic constrictive pericarditis is complete **surgical resection of the pericardium.**

restrictive

- **HX:** storage and infiltrative diseases
- **PE:** may be kussmaul sign
 - Prominent apical impulse
 - S3 rarely S4
 - Regurgitant Murmur is common
- **ECG:** Left axis deviation
 - Atrial fib.
- **Echo:** Increased wall thickness
 - Thickened valves
- **CT/MRI:** Pericardium usually normal
 - <2mm
- **Endomyocardial bx:** reveal specific disease

constrictive

- **HX:** TB, trauma...
- **PE:** Kussmaul sign usually +ve
 - Apical impulse not palpable
 - Pericardial knock
 - Regurgitant murmur uncommon
- **ECG:** Square root or Dip plateau sign
- **Echo:** normal wall thickness
 - Pericardial thickening may be
- **CT/ MRI**
 - Pericardium may be thickened >4mm
- **Endomyocardial bx:** may be normal

Cardiac tamponade

- Low cardiac output state
- NO Kussmaul's sign
- RA: blunted y descent
- Decreased heart sounds
- Pulsus Paradoxus
- Echo effusion

Constrictive

- Low cardiac output state
- Kussmaul's sign
- RA: rapid y descent
- Pericardial "knock"
- No pulsus paradoxus
- Echo thick pericardium

Cardiac Tamponade

- Important associated symptoms:
 - Dyspnea
 - Chest Pain
 - Feeling of “Fullness”
 - Nausea/Abdominal pain (from hepatic/visceral congestion)
 - Dysphagia (from esophageal compression)
- Nonspecific Sxs:
 - Lethargy, fever, cough, weakness, fatigue, anorexia, palpitations
- Classically tachycardic, hypotensive and tachypenic
 - BUT pts with pre-existing HTN may still be hypertensive with tamponade!
- Diminished heart sounds
- Kussmaul sign: inspiratory elevation of JVP
 - Present only in underlying constrictive disease
- Possible friction rub

Cont'd

- Classic “**Beck’s Triad**”
 - Decreased arterial BP
 - Increased JVP
 - Quiet heart sounds
- Pulsus Paradoxus
 - exaggeration of the normal inspiratory decrease in BP (ie, it is present normally)
 - “**Abnormal**” pulsus paradoxus = >10mmHg difference
 - A pulsus paradoxus >20 mm Hg in a child with pericarditis is an **indicator of the presence of cardiac tamponade**
 - a 10–20 mm Hg change is equivocal.

Cont'd

- Ddx (pulsus paradox)
- Severe COPD
- CHF
- Mitral Stenosis
- Massive PE
- Severe hypovolemic shock
- Obesity
- Tense ascites
- Conditions that can mask wide pulsus paradoxus:
 - Hypotension
 - Pericardial adhesions
 - Aortic regurg
 - ASD
 - RVH

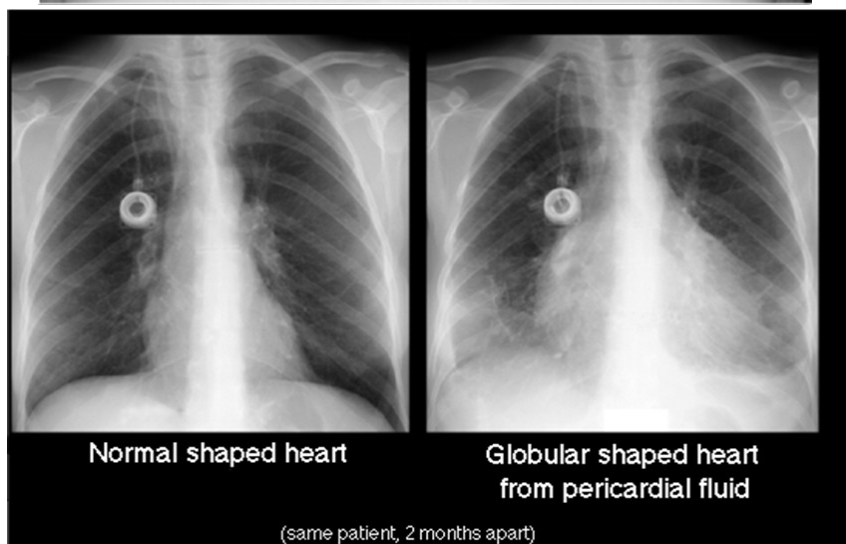
DX

1. Cardiac Catheterization
2. Chest Radiography:
3. Electrocardiogram:
 - Electrical alternans
 - Every other QRS is of smaller voltage
 - Atrial arrhythmias
4. ECHO:
 - Circumferential fluid layer
 - **Specific sign** is chamber collapse:

Cont'd



- CXR
- Depends on the size of the effusion
 - Enlarged, **globular heart** (“water bottle”) or flask shaped
 - Epicardial fat stripe
 - Can be normal



Cont'd

- MX

1. **Pericardiocentesis:**

Blindly in the case of an emergency

With ECHO, fluoroscopy or CT guidance

2. **Pericardiotomy:**

If the heart cannot be reached by a needle/catheter.

Indicated in patients with intrapericardial bleeding, clotted hemopericardium.

3. **Positive airway pressure**

should be **avoided** as it decreases cardiac output

Pericardial Effusions

- Secondary to many processes (viral, Tb, malignancies, uremia)
- TYPES :
 - **Transudative** (CHF, Myxoedema, Nephrotic Syndrome),
 - **Exudative** (tuberculosis, spread from empyema)
 - **Haemorrhagic** (trauma, rupture of aneurysms, malignant effusion).

CHF (mx principle)

1. supportive

- Bed rest, frequent positioning, DVT prophylaxis
- salt restriction
- Water restriction
- Oxygen
- Nutrition
- Daily weight

2. RX of the failed heart

- Overload reducers
 - ACE I
 - B blockers
 - Spironolactone
- Preload reducers
 - diuretics
- Increase myocardial contractility
 - Digoxin
 - Dopamin
 - dobutamin

Cont'd

3. RX of the underlying condition

- Cardiac surgery
- Correct rhythm abnormality
- Transfuse if anemia
- Anti hypertensive
- Correct met. abnormality

4. Rx of the precipitating factors

- Hypertension
- Endocarditis
- Arrhythmia
- Rheumatic fever
- Thyrotoxicosis
- Fever
- Arrhythmia
- Infection
- Liver
- Salt/ stress